

# DIVYA GAVANE

Semiconductor Engineer | Custom IC Design | Memory | TCAD | Advanced Packaging

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## PROFESSIONAL SUMMARY

Semiconductor engineer and NUS M.S. Nanoelectronics candidate with three years of Micron experience spanning NAND/mNAND package hardware, reliability, manufacturability and design-flow improvement. Hands-on researcher who designed, integrated, fabricated and measured a 32-bit x 8-vector FeTFT BCAM macro. Brings a device-to-system view across custom IC design, GDS layout, Sentaurus TCAD, CMOS/SRAM, RTL, STA and PPA analysis.

## EDUCATION

### National University of Singapore | Aug 2025 - present

M.S. in Electrical Engineering (Nanoelectronics) | CGPA: 4.33 / 5

### MKSSS's Cummins College of Engineering | Aug 2018 - May 2022

B.Tech in Electronics and Telecommunications | CGPA: 9.09 / 10

## GRADUATE RESEARCH

### Monolithic 3D-stacked ZnO TFT / Fe-ZnO FeTFT BCAM for Hamming-distance search

National University of Singapore | Jan 2026 - present

- Used measured I-V data from fabricated 2T-2FeTFT bitcells to establish HVT/LVT behaviour, circuit biasing and peripheral-design requirements.
- Designed transistor-level schematics and custom GDS layouts for decoders, drivers, precharge/equalizer circuits, differential multiplexers, logic cells and a sense amplifier.
- Integrated the array and peripherals into a 32-bit x 8-vector, 256-cell macro supporting write, read and Hamming-distance search; had the complete macro fabricated.
- Performed post-fabrication electrical measurements to validate write/read functionality and characterise Hamming-distance search behaviour.
- Contributed GDS macros, SPICE netlists, testbenches and HSPICE/Verilog-A simulation assets to the open-source 3D\_TFT repository.

## INDUSTRY EXPERIENCE

### Micron Technology, Hyderabad | Package Hardware Engineer | Jun 2024 - Jun 2025

- Established a package-schematic design and review flow with cross-functional stakeholders, tracking reviews and approvals in JIRA.
- Standardised package-library naming and component classification to improve identification and reuse across schematics and layouts.
- Developed NAND/mNAND package schematics; schematic-layout co-design reduced the overall package-design cycle time by 10%.
- Authored technical learning submissions on FBLR autorouting and schematic-layout co-design; proposed a wire-bond import/export improvement for patent review.

### Micron Technology | Associate Package Design Engineer | Aug 2022 - May 2024

- Designed package QA boards for JEDEC B111A, JESD22-B111, JESD22-A104/9701A, JESD22-B112/9702 and AEC-Q007 reliability requirements.
- Developed PCB single units and substrate layouts to meet SIRD requirements, balancing electrical performance and manufacturability.
- Won the April 2023 PDE Hackathon with a study on substrate trace cracking and prevention.

## DIVYA GAVANE | SELECTED PROJECTS & SKILLS

### ACADEMIC PROJECTS

#### PN-junction virtual fabrication and TCAD characterisation | Mar 2026

Sentaurus Workbench, SProcess, SDevice, SVisual

- Co-developed a 2D process flow covering STI, well and P+/N+ implants, anneals, contact etch, aluminium metallisation and SDevice handoff.
- Automated TDR checkpoints and junction-depth extraction. Sweeping acceptor doping from  $1.1 \times 10^{14}$  to  $1.1 \times 10^{16} \text{ cm}^{-3}$  reduced junction depth from 2.526 to 1.816 micrometres.
- Analysed equilibrium and biased band diagrams and I-V behaviour, including high-level injection and series-resistance effects.

#### RTL design, verification and synthesis of RISC control and FP8 arithmetic | Apr 2026

Verilog, Synopsys VCS, Design Compiler, Tcl, Shell

- Implemented and verified a 16-bit RISC program-counter block; automated synthesis and analysed critical paths, setup/hold timing and area.
- Built and regression-tested 3-, 4- and 5-stage FP8 adders using 256 test vectors. Clock sweeps and PPA extraction reduced the minimum feasible period from 1.0 ns to 0.6 ns.
- Selected the 4-stage design as the best trade-off: 1.43x throughput with 45.74% additional area.

#### Energy-efficient CMOS, sequential logic and SRAM stability | Oct 2025

Cadence Virtuoso

- Sized multistage logic paths using logical effort and energy-delay analysis; designed a transmission-gate flip-flop and extracted setup and clock-to-Q timing.
- Characterised leakage and body-bias effects and ran 1,000 Monte Carlo simulations for delay variability.
- Extracted 45-nm 6T SRAM hold/read/write noise margins of 124.5 / 102.6 / 435.0 mV; identified read stability as the limiting condition.

#### 45-nm standard cells and post-layout characterisation | Nov 2025

Cadence Virtuoso, Spectre, FreePDK45

- Built DRC/LVS-clean inverter, NAND2 and NOR2 layouts; characterised delay, power, FO4 delay and logical effort.
- Modelled interconnects with pi-RC networks and Elmore delay; built a ring oscillator and quantified parasitic penalties after extraction.

#### PN junction, MOS capacitor and MOSFET simulation | Sep 2025

Sentaurus SDE, SDevice

- Built device structures and analysed band diagrams and carrier transport. Extracted threshold voltage, transconductance and subthreshold swing; studied doping, oxide thickness, gate work function and temperature.

### TOOLS & SKILLS

**Circuit:** Cadence Virtuoso, Spectre, HSPICE, Verilog-A, GDS layout, DRC/LVS, Monte Carlo.

**Digital:** Verilog, Synopsys VCS, Design Compiler, Tcl, Shell, RTL, synthesis, STA, PPA.

**Devices:** Sentaurus Workbench, SProcess, SDevice, SVisual; process integration and I-V analysis.

**Packaging:** Cadence package, PCB and substrate design, FBLR autorouting, JEDEC/AEC-Q boards, JIRA.